VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

A project report
on

"Design of MIMO Antenna System for 5G Mobile Communication"

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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PES INSTITUTE OF TECHNOLOGY AND MANAGEMENT

NH-206, Sagar Road, Shivamogga – 577 204

2025–2026

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To prepare students for professional accomplishments and responsible global citizenship while fostering continuous learning and to provide state-of-the-art education through the committed and highly skilled faculty by partnering and collaborating with industry and R&D institutes.

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PEO 2: To give exposures to emerging technologies, adequate training and opportunities to work as team on multidisciplinary projects with effective communication skills.

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Certificate from the Institute

This is to certify that the project work was carried out jointly by

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are bonafide students of the Bachelor of Electronics and Communication Engineering program of the Institute (2023–2025 Batch), affiliated to Visvesvaraya Technological University, Belagavi.

A project report entitled **"Design of MIMO Antenna System for 5G Mobile Communication"** has been submitted by them under the guidance of Mrs.Sangeetha B S, Assistant Professor, Department of Electronics and Communication Engineering , PESITM, Shivamogga, in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering.

Mr.Hanumanthappa N Magalada	**Dr. Vishnu V M**
Assistant Professor	**Professor and Head**

Viva-voce Examination Date: _____

Signature of Internal Examiner

Signature of External Examiner

DECLARATION

We, **NAREESH G P, SHIVAKANTAGOUDA J K, SUHAS M D** and **SHREYAS H M** hereby declare that the project report entitled **"Design of MIMO Antenna System for 5G Mobile Communication"** has been carried out by me under the guidance of Mr.Hanumanthappa N Magalada, Assistant Professor, Department of ECE, PESITM, Shivamogga.

I further declare that this project report is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi.

I also declare that this report is based on the original work undertaken by me and has not been submitted for the award of any degree or diploma from any other University/Institution.

Place: Shivamogga
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Abstract

The rapid growth of wireless communication systems has increased the demand for compact, efficient, and high-speed antennas. Multiple Input Multiple Output (MIMO) technology is widely used in modern communication systems to improve data rate, signal quality, and reliability. This project titled **"Design of MIMO Antenna System for 5G Mobile Communication".**presents the design and fabrication of a compact MIMO antenna suitable for modern wireless and 5G applications.

The antenna is designed using ANSYS HFSS software and fabricated on an FR4 substrate. The proposed antenna consists of two rectangular microstrip patch elements with microstrip feed lines and a full copper ground plane. Proper spacing is maintained between the antenna elements to reduce mutual coupling and improve isolation.

The antenna performance is analyzed using parameters such as return loss (S11), Voltage Standing Wave Ratio (VSWR), gain, radiation pattern, and isolation. The fabricated antenna is tested using a Vector Network Analyzer (VNA). The designed antenna provides compact size, low cost, improved signal quality, and suitability for 5G communication systems.

The proposed system offers advantages such as better bandwidth, reduced signal fading, enhanced communication reliability, and easy fabrication. Therefore, the designed antenna is highly suitable for wireless communication, Wi-Fi, IoT, and future 5G applications.

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Chapter 1

Introduction

Wireless communication systems require compact and efficient antennas for high-speed data transmission and reliable communication. Microstrip patch antennas are widely used because of their small size, low cost, and easy fabrication. However, single antennas suffer from low bandwidth and signal fading. To overcome these problems, MIMO (Multiple Input Mul- tiple Output) technology is used. In this project, a 2×2 MIMO microstrip patch antenna operating at 3.5 GHz is designed using ANSYS HFSS soft- ware and fabricated on an FR4 substrate. The antenna improves signal quality, bandwidth, and communication performance, making it suitable for wireless and 5G applications

1.1 Problem Statement

Modern wireless communication systems require high data speed, better signal quality, compact antenna size, and reliable communication. Conventional single antennas suffer from low bandwidth, signal fading, and reduced efficiency in high-speed communication systems. Hence, there is a need for a compact and efficient MIMO microstrip antenna.

1.2 Project Description

The project focuses on the design and fabrication of a 2 × 2 MIMO microstrip patch antenna operating at 3.5 GHz for wireless communication applications. The antenna is designed using HFSS software and fabricated on an FR4 substrate. The project includes simulation, PCB fabrication, SMA connector integration, and testing using a Vector

Network Analyzer (VNA). The designed antenna improves signal quality, bandwidth, and communication reliability.

1.3 Objectives

- To achieve antenna operation at 3.5 GHz frequency suitable for 5G communication.
- To improve bandwidth, gain, and signal quality using MIMO tech- nology.
- To reduce mutual coupling between antenna elements for better iso- lation.
- To analyze antenna parameters such as return loss (S11), VSWR, gain, efficiency, and radiation pattern.
- To simulate and optimize the antenna design using ANSYS HFSS software.
- To fabricate the antenna using FR4 substrate and PCB technology.

1.4 Scope

The scope of this project is to design, simulate, fabricate, and test a compact 2×2 MIMO microstrip patch antenna operating at 3.5 GHz for wireless communication applications. The antenna is designed us- ing HFSS software and fabricated on an FR4 substrate using PCB fabrication techniques. The project mainly focuses on improving an- tenna performance parameters such as return loss, bandwidth, gain, VSWR, and isolation between antenna elements. The fabricated an- tenna is tested using a Vector Network Analyzer (VNA) to verify the simulated results. This project can be applied in modern wireless systems such as 5G, Wi-Fi, and IoT communication devices, and it can be further extended for advanced multi-band and high-frequency communication applications.

1.5 Purpose

The purpose of this project is to design and develop a compact 2×2 MIMO microstrip patch antenna operating at 3.5 GHz for wireless communication applications. The project aims to improve bandwidth, signal quality, data transmission speed, and communication

reliabil- ity using MIMO technology. The antenna is designed using ANSYS HFSS software and fabricated on an FR4 substrate for applications such as 5G, Wi-Fi, LTE, and IoT communication systems.

1.6 Relevance to Sustainable Development Goals (SDGs)

This project supports Sustainable Development Goals by improving communication technology and enabling better digital connectivity. Relevant SDGs:

SDG 9 – Industry, Innovation and Infrastructure Promotes advanced communication technology and innovation.

SDG 4 – Quality Education Enhances practical learning in antenna design, simulation, and fabrication.

SDG 11 – Sustainable Cities and Communities Supports smart com- munication systems for modern infrastructure.

Chapter 2

Literature Review / Existing System

With the rapid growth of wireless communication systems and increasing demand for high-speed data transmission, Multiple Input Multiple Output (MIMO) antenna technology has become one of the most important components in modern 5G communication systems. MIMO antennas improve channel capacity, increase spectral efficiency, reduce multipath fading, and provide reliable communication. Researchers are continuously developing compact, high-gain, wideband, and low-correlation MIMO antenna systems suitable for sub-6 GHz and millimeter-wave 5G applications.

[**1**] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed., Wiley Publications, 2016.

Balanis explains the fundamental concepts of antenna theory including radiation pattern, gain, directivity, polarization, impedance matching, and antenna arrays. The book provides detailed knowledge about microstrip patch antennas which are widely used in MIMO communication systems due to their compact size, low profile, and easy fabrication. Different feeding techniques and antenna parameters are also discussed. This reference forms the theoretical foundation for understanding MIMO antenna design and analysis.

[**2**] David M. Pozar, *Microwave Engineering*, 4th ed., Wiley Publications, 2012.

Pozar discusses microwave engineering principles such as transmission lines, scattering parameters, impedance matching, and electromagnetic wave propagation. These concepts are very important for designing high-frequency MIMO antennas used in modern wireless communication systems. The book also explains microwave circuit

analysis and antenna integration techniques required for 5G antenna development.

[**3**] M. M. Morsy, "Compact eight-element MIMO antenna array for sub-6 GHz mobile applications," *SN Applied Sciences*, vol. 5, 2023.

Morsy proposed a compact eight-element MIMO antenna array for sub-6 GHz mobile communication applications. The antenna system was specially designed for modern 5G smartphones and portable wireless devices. The proposed design achieved high isolation and low mutual coupling between antenna elements while maintaining compact dimensions. The study also reported good Envelope Correlation Coefficient (ECC) and diversity gain performance, making the antenna suitable for high-speed wireless communication systems.

[**4**] H. Alsaif and M. A. H. Eleiwa, "Compact design of 2×2 MIMO antenna with super-wide bandwidth for millimeter wavelength systems," *Symmetry*, vol. 13, no. 233, 2021.

Alsaif and Eleiwa designed a compact 2×2 MIMO antenna system with super-wide bandwidth suitable for millimeter-wave communication systems. The proposed antenna used defected ground structures and optimized patch dimensions to improve bandwidth and isolation characteristics. The antenna achieved stable radiation performance and supported multiple frequency bands used in 5G communication systems. The compact structure makes it suitable for handheld and wearable communication devices.

[**5**] Asadullah et al., "Dual-band composite high gain MIMO antenna for 5G NR applications," *Scientific Reports*, vol. 16, 2026.

Asadullah and co-authors proposed a dual-band composite high-gain MIMO antenna designed for 5G New Radio applications. The antenna supported dual-band operation with improved gain and radiation efficiency. The study focused on reducing mutual coupling between antenna elements while maintaining compact dimensions. The proposed antenna achieved wide bandwidth and stable radiation characteristics suitable for future wireless communication systems.

[**6**] M. Ikram, "5G/B5G Internet of Things MIMO Antenna Design," *Signals*, vol. 3, pp. 29–37, 2022.

Ikram discussed the role of MIMO antenna technology in 5G and Beyond 5G Internet of Things applications. The paper explained how MIMO systems improve communication reliability, increase network capacity, and support large numbers of connected devices. Different compact antenna structures suitable for IoT systems were analyzed. The study also highlighted challenges such as limited space, interference reduction, and low power consumption.

[**7**] D. Wang et al., "An overview of transmission theory and techniques of large-scale antenna systems for 5G wireless communications," *arXiv preprint arXiv:1605.03426*, 2016.

Wang and co-authors provided an overview of large-scale antenna systems and transmission techniques used in 5G wireless communication systems. The paper explained beamforming, channel estimation, spatial multiplexing, and massive MIMO technology. Large-scale antenna systems improve spectral efficiency and communication reliability. The study also discussed practical challenges such as hardware complexity and power consumption.

[**8**] T. Raj, R. Mishra, P. Kumar, and A. Kapoor, "Advances in MIMO antenna design for 5G: A comprehensive review," *Sensors*, vol. 23, 6329, 2023.

Raj et al. reviewed recent advances in MIMO antenna design for 5G communication systems. The paper discussed techniques such as Defected Ground Structures (DGS), Electromagnetic Band Gap (EBG), parasitic elements, and neutralization lines used for isolation improvement. The study emphasized the importance of compact size, high gain, low ECC, and wide bandwidth in modern antenna systems.

[**9**] W. T. Sethi et al., "Pattern diversity based four-element dual-band MIMO patch antenna for 5G mmWave communication networks," *Journal of Infrared, Millimeter, and Terahertz Waves*, vol. 45, pp. 521–537, 2024.

Sethi and co-authors proposed a pattern diversity-based four-element dual-band MIMO patch antenna for millimeter-wave 5G communication networks. The antenna achieved high isolation, stable radiation patterns, and improved diversity performance. Pattern diversity helped in reducing signal fading and improving communication reliability. The proposed antenna was suitable for high-speed wireless communication systems.

[**10**] H. Hambergerami, R. Kazemi, and A. E. Fathy, "Development of a metasurface-based slot antenna for 5G MIMO applications," *Scientific Reports*, vol. 14, 8016, 2024.

Hambergerami and co-authors developed a metasurface-based slot antenna for 5G MIMO applications.

2.1 Introduction

Recent studies show that modern wireless communication systems require high data rates, low latency, and reliable connectivity to support 5G mobile applications such as IoT, smart devices, autonomous systems, and high-speed internet services [**?**]. Traditional antenna systems face challenges like low bandwidth, signal fading, and limited channel capacity in high-frequency communication environments

Research on 5G technology highlights the importance of Multiple Input Multiple Output (MIMO) antenna systems, which use multiple transmitting and receiving antennas to improve communication performance, increase data throughput, and reduce signal interference
Advanced antenna designs such as microstrip patch antennas are widely preferred for 5G applications because of their compact size, low profile, lightweight structure, and ease of fabrication .

Existing Systems

- **Single Antenna Systems:** Traditional communication systems use single antennas, which provide limited bandwidth and lower data transmission rates.
- **Conventional Microstrip Antennas:** Existing microstrip antennas are compact but suffer from narrow bandwidth and reduced gain for high-frequency 5G applications.
- **Basic MIMO Systems:** Some MIMO antenna systems improve channel capacity but experience high mutual coupling and poor isolation between antenna elements.
- **Millimeter Wave Antenna Systems:** Certain 5G antennas operate at millimeter-wave frequencies but require complex fabrication techniques and high design accuracy.
- **Compact Smartphone Antennas:** Existing mobile antennas support 4G and partial 5G communication but often face challenges in achieving high efficiency and multi-band operation.

Limitations of the Existing Systems

- Limited bandwidth for ultra-high-speed 5G communication.
- High mutual coupling between antenna elements.
- Reduced isolation and efficiency in compact antenna structures.
- Difficulty in maintaining stable performance at millimeter-wave frequencies.
- Increased signal loss and interference in dense communication environments.
- Complex fabrication and higher implementation cost.

Conclusion

The proposed **Design of MIMO Antenna System for 5G Mobile Communication** provides an efficient solution for modern wireless communication requirements. By using multiple antenna elements and advanced isolation techniques, the system improves bandwidth, gain, efficiency, and overall communication performance. The proposed MIMO antenna design is compact, cost-effective, and highly suitable for future 5G mobile communication applications.

2.2 Existing Work

Microstrip antennas and MIMO technology are widely used in wire- less communication systems due to their compact size and improved performance. Many researchers have designed antennas for Wi-Fi, LTE, and 5G applications. Existing systems mainly focus on improv- ing bandwidth, gain, and reducing mutual coupling between antenna elements. HFSS and CST software tools are commonly used for an- tenna simulation and optimization. FR4 substrate is widely preferred because of its low cost and easy availability.

2.3 Problems with the Existing Systems

- **Traditional Mechanical Locker Systems:** Most bank lockers use physical keys and manual verification methods, which are vulnerable to key duplication,

theft, and unauthorized access.

- **Password-Based Security Systems:** Some locker systems use passwords or PIN-based access, but they provide only single-level security and can be easily guessed or shared.
- **Biometric Locker Systems:** Fingerprint and face recognition systems improve security, but many existing systems use only one biometric method without additional authentication layers.
- **IoT-Based Monitoring Systems:** Certain smart locker systems use IoT technology for remote monitoring and alert generation, but they lack intelligent AI-based threat detection and analysis.
- **GSM Alert Systems:** GSM-based locker systems can send SMS alerts during unauthorized access attempts, but they do not provide complete multi-level authentication and real-time cloud monitoring.

2.4 Gap Identification

Microstrip patch antennas with MIMO technology are widely used in wireless communication systems due to their compact size, low cost, and better performance. Many researchers have designed antennas for 5G, Wi-Fi, and LTE applications focusing on improving bandwidth, gain, and isolation. ANSYS HFSS is commonly used for antenna simulation and analysis.

Chapter 3

Methodology

The 2×2 MIMO microstrip patch antenna is designed using ANSYS HFSS software for 3.5 GHz operation on an FR4 substrate. The antenna is simulated to analyze parameters such as S11, VSWR, gain, and radiation pattern. After optimization, the design is fabricated using PCB technology and tested using a Vector Network Analyzer (VNA) to verify the antenna performance

3.1 Requirement Analysis

The project requires both software and hardware components for antenna design, fabrication, and testing.

3.1.1 Functional Requirements

- **Operate at 3.5 GHz frequency.**
- **Support MIMO communication.**
- **Provide good return loss (S11 ¡ -10 dB).**
- **Maintain low VSWR.**
- **Reduce mutual coupling.**
- **Support signal transmission and reception.**

3.1.2 Non-Functional Requirements

- **Compact in size.**
- **Low cost..**
- **Easy to fabricate.**

Table 3.1: Requirement Specification

Requirement Type	Description
Functional	Operate efficiently at 3.5 GHz frequency for wireless and 5G communication applications.
Functional	Support MIMO communication to improve bandwidth, signal quality, and data transmission speed.
Non-functional	Compact and lightweight antenna design for easy integration into wireless devices.
Non-functional	Reliable, low-cost, and easy-to-fabricate antenna system using FR4 substrate.

3.2 System Design

The antenna is designed using HFSS software a 2×2 MIMO structure is created using two microstrip patch antenna elements placed on an FR4 substrate. Microstrip feed lines are used to excite the antenna elements. Proper spacing is maintained between the patches to reduce mutual coupling and improve isolation between antenna elements.

The proposed antenna system operates at 5G frequency bands and provides improved bandwidth, gain, radiation efficiency, and signal reliability. Techniques such as Defected Ground Structure (DGS) and slot loading are used to enhance antenna performance and reduce interference.

The above figure shows the structure of the proposed 2×2 MIMO microstrip patch antenna system designed for 5G mobile communication applications. The antenna elements are arranged symmetrically on the substrate to achieve better isolation and efficient signal transmission.

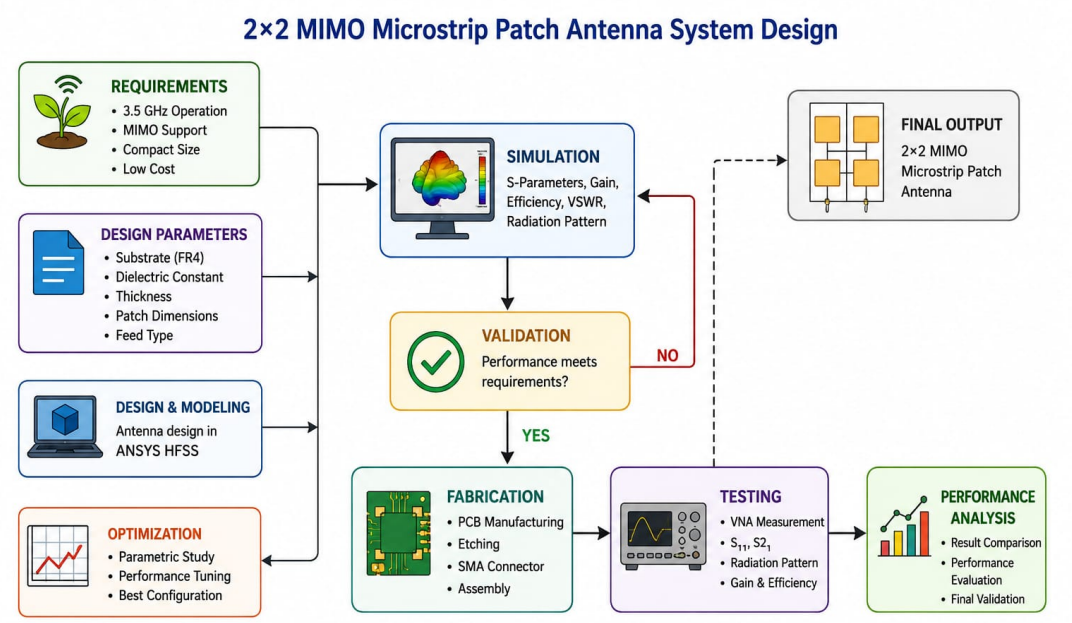


Figure 3.1: 2×2 MIMO Microstrip Patch Antenna System Design

3.2.1 System Architecture

The proposed MIMO antenna system for 5G mobile communication consists of multiple microstrip patch antenna elements arranged in a compact structure to improve bandwidth, gain, and signal reliability. The system is designed to operate at 5G frequency bands with reduced mutual coupling and improved isolation between antenna elements.

The architecture includes a substrate material, radiating patch elements, feeding network, ground plane, and defected ground structure (DGS). Advanced techniques such as slot loading and parasitic elements are used to enhance antenna performance and reduce interference.

The transmitting antenna elements send multiple data streams simultaneously, while the receiving antenna elements capture the signals efficiently to improve channel capacity and communication quality. The overall system supports high-speed wireless communication with better radiation efficiency and stable performance for 5G mobile applications.

3.3 Algorithms Used

- **MIMO Signal Processing Algorithm:** Used to improve channel capacity and data transmission efficiency by transmitting multiple signals simultaneously.
- **Beamforming Algorithm:** Helps direct antenna radiation toward the desired user to improve signal strength and reduce interference.
- **Mutual Coupling Reduction Technique:** Uses Defected Ground Structure (DGS) and slot techniques to improve isolation between antenna elements.
- **Optimization Algorithm:** Parameters such as antenna dimensions, spacing, and frequency are optimized for better bandwidth and gain performance.
- **Radiation Pattern Analysis:** Used to analyze antenna directivity, gain, and radiation efficiency for 5G communication.

3.4 Tools and Technologies

- **HFSS (High Frequency Structure Simulator):** Used for antenna design and electromagnetic simulation.
- **CST Microwave Studio:** Used to analyze antenna parameters such as gain, return loss, and radiation pattern.
- **MATLAB:** Used for signal processing, graph plotting, and performance analysis.
- **Microstrip Patch Antenna Technology:** Used for compact and lightweight antenna structure design.
- **Defected Ground Structure (DGS):** Used to reduce mutual coupling and improve isolation.
- **5G Wireless Communication Technology:** Supports high-speed data transmission and low latency communication.
- **FR4 / Rogers RT Duroid Substrate:** Used as dielectric substrate material for antenna fabrication.

Chapter 4

Implementation and Testing

4.1 System Implementation

System implementation is the process of converting the proposed design into a working model using suitable software and design techniques. In this project, a MIMO antenna system for 5G mobile communication is designed and implemented using ANSYS HFSS (High-Frequency Structure Simulator).

ANSYS HFSS is an electromagnetic simulation software used to design and analyse high-frequency electronic components, including antennas. It helps evaluate antenna characteristics such as return loss, impedance matching, frequency response and radiation behaviour.

The proposed MIMO antenna system consists of multiple antenna elements arranged to improve wireless communication performance. The antenna structure is designed by considering geometrical dimensions, feeding arrangements, substrate thickness and spacing between antenna elements.

Sl. No.	Parameter	Description	Dimension
1	D12 = D34	Outer antenna dimension	42.5 mm
2	d12 = d34	Inner antenna dimension	18.5 mm
3	d23	Inner spacing between antenna elements	10 mm
4	D23	Outer spacing between antenna elements	32.5 mm
5	L1	Antenna length	22 mm
6	h	Substrate thickness	1.5 mm
7	D	Overlap length	10 mm
8	W	Antenna width	1 mm
9	r_o	Outer coaxial probe radius	0.5 mm
10	r_i	Inner coaxial probe radius	0.25 mm
11	L_p	Coaxial probe length	5 mm
12	W_f	Feedline width	1 mm
13	L_f	Feedline length	10.2 mm

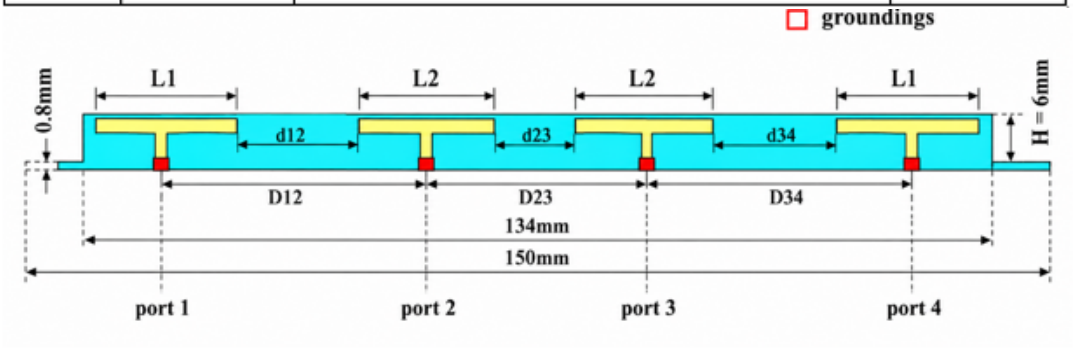


Figure 4.1: Design Parameters of Proposed MIMO Antenna

4.2 System Testing

System testing is the process of verifying whether the implemented system performs according to the expected design requirements.

In this project, system testing is performed through electromagnetic simulations using ANSYS HFSS. The simulated results are analysed to understand the performance of the proposed MIMO antenna system.

The testing process focuses on the following characteristics:

- Frequency response of the antenna.
- Return loss and impedance matching.
- Wave-port output characteristics.
- Standing Wave Ratio (SWR).
- Interaction between antenna elements.
- Overall behaviour of the MIMO antenna system.

The simulation results help identify possible design problems and determine whether the antenna structure provides the required electromagnetic response.

4.2.1 Unit Testing

Unit testing is the process of testing individual components of a system separately to ensure that each component functions correctly.

In the proposed MIMO antenna system, unit testing is carried out by examining the individual antenna elements and their corresponding feeding arrangements.

The following tests are performed:

- Antenna Element Testing: Each antenna element is examined to verify its geometrical structure and simulated electromagnetic response.
- Feedline Testing: The feeding arrangement is checked to ensure proper excitation of the antenna elements.

- Wave-Port Testing: Individual wave-port outputs are observed to study the response of the antenna structure.
- Frequency Response Testing: The frequency response is analysed to identify the operating frequency regions.
- SWR Testing: The standing wave ratio is examined to evaluate the impedance matching behaviour.

4.2.2 Integration Testing

Integration testing is performed to verify the interaction and combined operation of different components of a system.

In this project, integration testing is carried out by analysing the complete MIMO antenna arrangement in ANSYS HFSS.

The following activities are performed:

- All designed antenna elements are considered together as a complete antenna system.
- The feeding arrangements are examined in the integrated structure.
- The wave-port outputs are analysed to understand the combined antenna response.
- The interaction between different antenna elements is studied.
- The combined SWR characteristics are examined.
- The overall simulated performance is evaluated.

Integration testing helps determine whether the individual antenna elements work together as a complete MIMO antenna system.

The results obtained from this testing process are used for further analysis and performance evaluation.

Chapter 5

Results and Discussion

5.1 Output Screenshots with Explanations

The proposed MIMO antenna system is simulated using ANSYS HFSS.
The simulation results are analysed to understand the electromagnetic behaviour and performance of the antenna.

The obtained outputs include the wave-port response, individual antenna outputs and combined SWR characteristics.

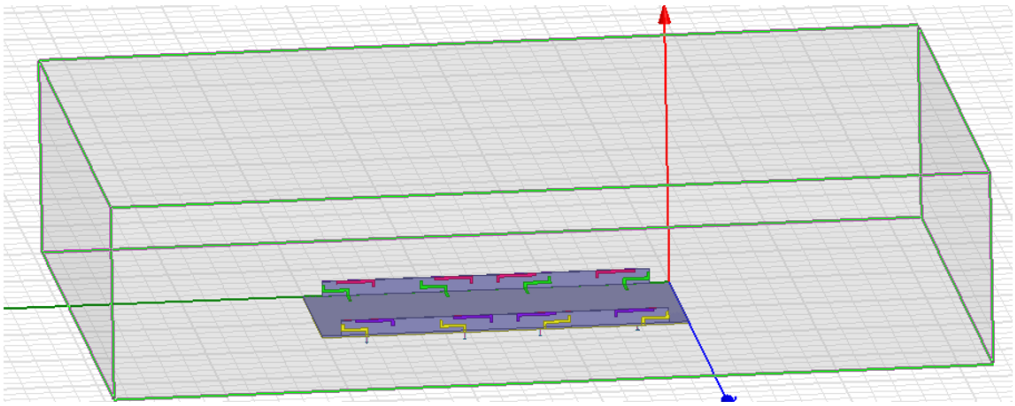


Figure 5.1: Simulated MIMO Antenna Structure in ANSYS HFSS.

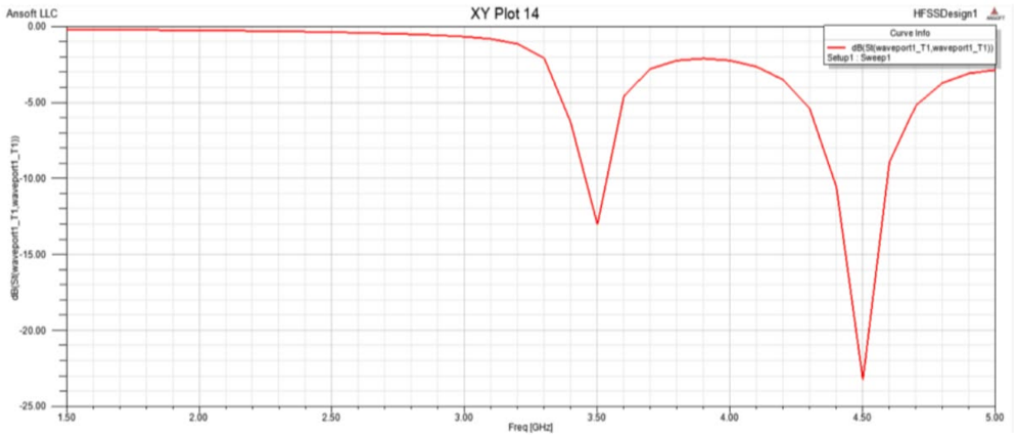


Figure 5.2: Return Loss (S11) Response of Wave Port 1

The wave-port 1 output represents the simulated frequency response of the antenna. The obtained graph shows a dual-band response, indicating two operating frequency regions in the simulated design.

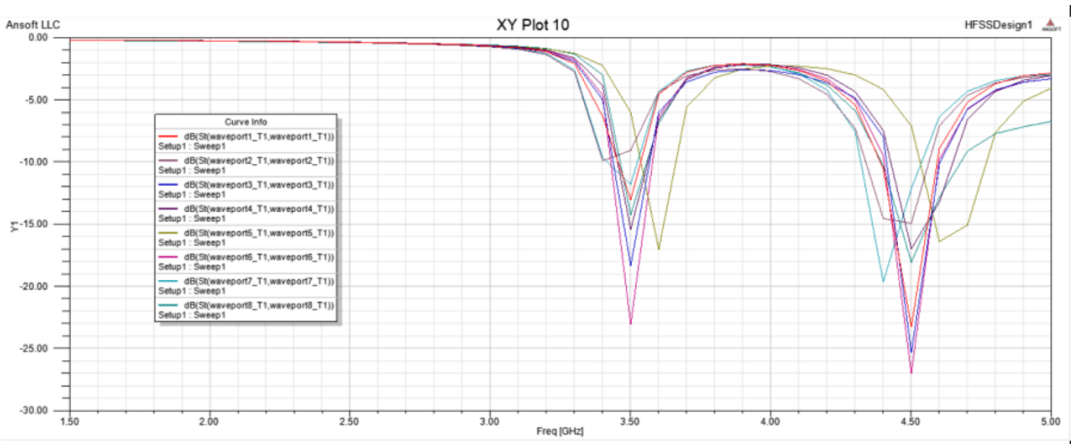


Figure 5.3: Return Loss Responses of Multiple Antenna Ports

The Standing Wave Ratio (SWR) graph represents the impedance matching behaviour of the antenna system. A lower SWR value indicates better impedance matching and reduced reflected power.

The SWR results are used to evaluate the feeding performance of the proposed antenna structure.

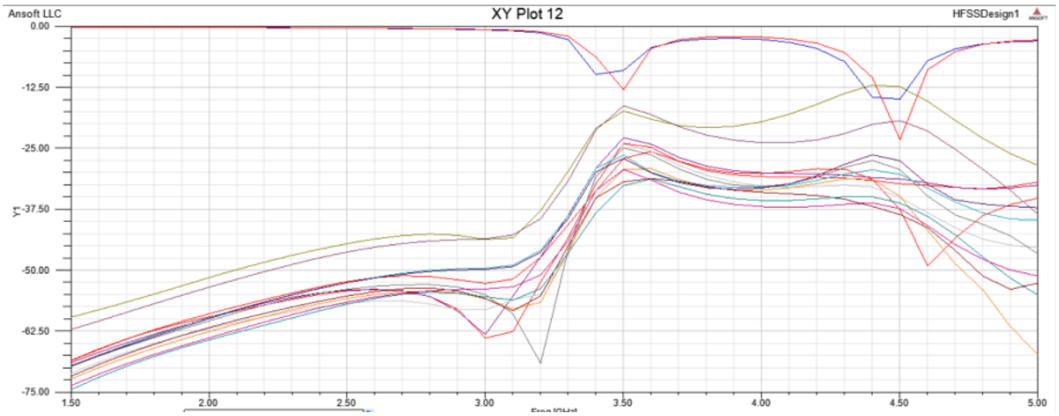


Figure 5.4: Combined SWR Output of the Antenna System.

5.2 Comparison with Existing Systems

MIMO antenna technology is widely used in modern wireless communication systems to improve data transmission capacity and communication reliability.

Conventional single-antenna systems use one antenna element for transmission or reception, whereas MIMO systems use multiple antenna elements to support multiple communication paths.

The proposed antenna system is designed with multiple antenna elements and a dual-band response

The proposed MIMO antenna system provides a multi-element design suitable for investigating 5G communication applications. Its actual performance advantage must be established using measured or simulated numerical results.

5.3 Performance Evaluation

Performance evaluation is carried out to determine the effectiveness of the proposed MIMO antenna system.

The following parameters are considered:

1. Return Loss

Return loss indicates the amount of signal power reflected back due to impedance mismatch. A more negative S11 value generally indicates better impedance match

2. Impedance Bandwidth

Impedance bandwidth represents the frequency range over which the antenna maintains acceptable impedance matching.

3. Standing Wave Ratio (SWR)

SWR is used to evaluate impedance matching between the antenna and feeding system. An SWR value closer to 1 indicates better matching.

4. Mutual Coupling

Mutual coupling represents the electromagnetic interaction between nearby antenna elements. Lower mutual coupling is generally desirable in MIMO antenna systems.

5. Envelope Correlation Coefficient (ECC)

ECC measures the correlation between antenna elements. A lower ECC value is generally preferred for better diversity performance.

6. Diversity Gain

Diversity gain indicates the improvement in communication reliability obtained through multiple antenna elements.

The simulation results obtained from ANSYS HFSS are used to evaluate these parameters. The dual-band response of wave-port 1 and the SWR characteristics provide information about the antenna's simulated performance.

Chapter 6

Conclusion and Future Enhancements

6.1 Conclusion

The project focuses on the design and simulation of a MIMO antenna system for 5G mobile communication using ANSYS HFSS.

The antenna structure is developed by considering different geometrical parameters, substrate thickness, antenna spacing and feeding arrangements.

The proposed antenna system is analysed through electromagnetic simulation. The obtained results include the dual-band response of wave-port 1, individual outputs of antennas 1 to 8 and combined SWR characteristics.

The simulation study helps understand the electromagnetic behaviour, impedance matching and operation of the proposed multi-element antenna structure.

Thus, the project demonstrates the design and simulation of a MIMO antenna system intended for 5G mobile communication applications.

6.2 Limitations

Although the proposed MIMO antenna system is designed and simulated, certain limitations remain.

- The current work is primarily based on electromagnetic simulation using ANSYS HFSS.
- Practical fabrication and experimental validation are not included in the current work.
- Antenna performance may vary due to fabrication tolerances and material properties.
- Further optimization may be required to improve impedance matching.
- Mutual coupling and isolation between antenna elements require detailed numerical evaluation.
- Experimental analysis of antenna gain, radiation efficiency and radiation patterns can provide additional performance information.

6.3 Future Enhancements

The proposed MIMO antenna system can be further improved to achieve better performance in advanced wireless communication applications.

- **Antenna Optimization:** Antenna dimensions can be optimized to improve the overall electromagnetic performance.
- **Mutual Coupling Reduction:**Additional isolation techniques can be introduced to reduce electromagnetic interaction between antenna elements.
- **B**andwidth Enhancement: The antenna structure can be modified to achieve wider operating bandwidth.
- **P**ractical Implementation: The proposed antenna can be fabricated and tested using suitable measurement equipment.
- **A**dvanced Performance Analysis: Further analysis of antenna gain, radiation efficiency, ECC and diversity gain can be performed.

- **5**G and Future Communication: The antenna design can be further investigated for advanced 5G and future wireless communication applications.

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Appendix A

Figures/Tables/ Screen Shots

[Insert your appendix content here]

Appendix B

Publications

[Publication Details/Certicate/Patents]